

Before You Begin

- Do not modify the motor in any way.
- If any parts are missing or damaged, call AeroTech at 435-865-7100.
- Use only AeroTech RMS reload kits to refurbish an RMS motor.
- Do not interchange parts from different reload kits.
- Do not reuse any parts of the RMS reload kit.
- Save the reload kit plastic bag for the used reload kit parts. Dispose of bag and parts properly.

Hardware and Supplies Required

RMS 98mm aft closure
RMS 98 case (see motor specification label for size)
98mm forward seal disk
98mm plugged forward closure
-or-
98mm reload adapter system (also refer to RAS instructions)
Synco™ Super Lube™ or other grease
Elmer's Glue-All or Gorilla Glue (if specified in separate grain bonding instructions)
30 minute epoxy (if specified in separate grain bonding instructions)
Hobby knife
Wet wipes or damp paper towels

Preparation for Flight

1. Insert the coated end of a FirstFire™, Firestar™ or other initiator through the nozzle throat until it stops against the smoke charge grain.
2. Secure the initiator to the nozzle with a piece of masking tape.
3. Install the motor into the rocket's motor mount tube. Ensure that the motor is securely retained in the rocket by using positive mechanical means to prevent it from being ejected during recovery system deployment.
4. Prepare the rocket's recovery system and then launch the rocket in accordance with the Tripoli Rocketry Association (TRA) Safety Code and National Fire Protection Association (NFPA) Code 1127.

Post-Recovery Cleanup

NOTE: Perform motor clean-up as soon as possible after motor firing. Propellant and smoke charge residues become difficult to remove after 24 hours and can lead to corrosion of the metal parts. Place the spent motor components in the reload kit plastic bag and dispose of properly.

1. After the motor has cooled down, unthread and remove the forward and aft closures.
2. Remove the smoke charge assembly components from the forward closure and discard. Using wet wipes or damp paper towels, remove all smoke charge and propellant residue from the closures.

Post-Recovery Cleanup (cont'd)

3. Remove and discard the forward and aft o-rings from the motor case. Remove the liner, forward seal disk and nozzle from the casing by pushing on the nozzle end. Remove the forward seal disk from the liner, and remove and discard the forward seal disk o-ring. **DO NOT DISCARD THE FORWARD SEAL DISK!** Discard the nozzle and liner. Using wet wipes or damp paper towels, wipe the inside of the casing and the forward seal disk to remove all propellant residue.

4. Apply a light coat of grease to all threads and the inside of the motor case. Reassemble metal parts and store motor in a dry place.

First Aid

For a minor burn, apply a burn ointment. For a severe burn, immerse the burned area in ice water at once and see a physician as quickly as possible. In the unlikely event of oral ingestion of the propellant, induce vomiting and see a physician as quickly as possible. AeroTech composite propellant consists primarily of Ammonium Perchlorate and a rubber-like plastic elastomer.

Disposal

Damaged or defective reload kits should be returned to RCS.

Fire Safety

Tests show that the pyrotechnic components of RMS™ reload kits will not explode in fires and normally will not ignite unless subjected to direct flame and then will burn slowly. Use water to fight any fires in which AeroTech RMS™ reload kit pyrotechnic components may become involved: Direct the water at the AeroTech RMS™ reload kit pyrotechnic components to keep them below their 550 deg. F autoignition temperature. Foam and carbon dioxide fire extinguishers will NOT extinguish burning propellants of the type used in RMS™ reload kit pyrotechnic components. Keep reload kit pyrotechnic components away from flames, sources of heat and flammable materials.

Warranty

GENERAL: No warranty, either expressed or implied is made regarding AeroTech/RCS products, except for replacement or repair, at RCS's option. Only those products which are proven to be defective in manufacture within 90 days or one year (applicable as described below), from the date of original purchase qualify. In no case will RCS warranty a product more than five (5) years after the date of manufacture. Incidental or consequential damages are not covered. For repair or replacement under this warranty, please send a copy of your paid invoice or other proof of purchase, the product involved, and a brief letter describing your experience with the reported failure. Freight or other fees incurred to send items to RCS are non-refundable. Note: Your state may provide additional rights not covered by this warranty.

RMS RELOAD KITS: (18-98mm) Covered 1 year from date of purchase: Failure of delay to ignite or to remain lit (i.e., partially remaining unburned delay element), ejection of nozzle insert, missing or defective parts. NOT COVERED: Failure to ignite, any failure of recovery system deployment not the result of delay element extinguishment, reload kits not used in accordance with instructions or modified in any way, any reload kit assembled more than twelve hours before flight, any incidental or consequential failure or damage related to use in a cluster or in staged applications.



HIGH-POWER RMS™ Reloadable Motor System™

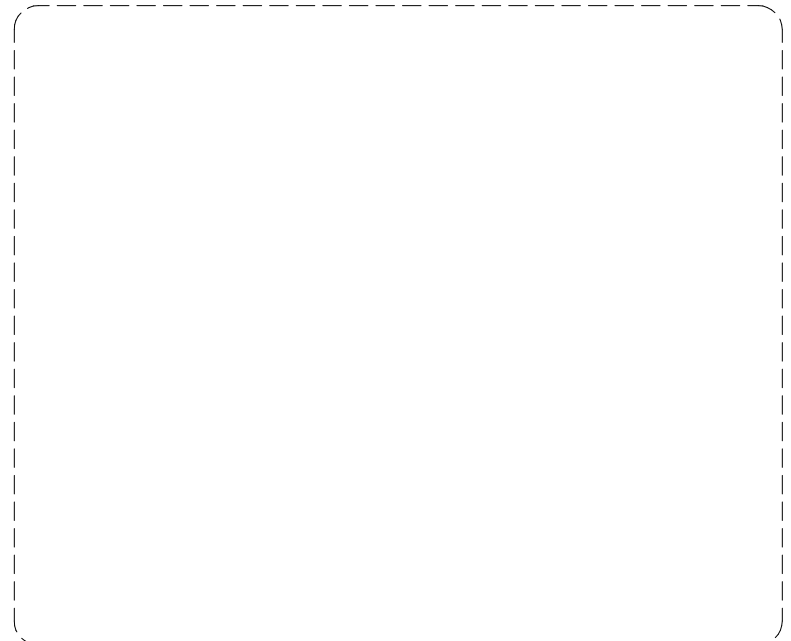
Rocket Motor Reload Kit For RMS-98/2560-20480 Motor Hardware

DO NOT OPEN RELOAD KIT UNTIL READY TO USE

NOTE: This reload kit MUST be used with motor hardware, separately packaged propellant grains and motor liner tube specified below. RMS-98 reload kits do not include an ejection charge. RMS-98 motors must be used in conjunction with a timer, altimeter or radio-actuated recovery system.

WARNING-FLAMMABLE: READ INSTRUCTIONS BEFORE USE. Use RMS reload kits only in accordance with instructions. Sale to persons under 18 years of age prohibited by federal law. For use only by certified users 18 years of age or older. Ignite by electrical means only. Do not smoke when loading RMS motors or use in the vicinity of open flames. **CAUTION:** Keep out of reach of children. Metalstorm™ propellants produce showers of hot sparks. Clear launch area of all combustible material for at least a 75 foot diameter radius around launcher. Follow NAR & TRA safety codes at all times. Motor hot after firing.

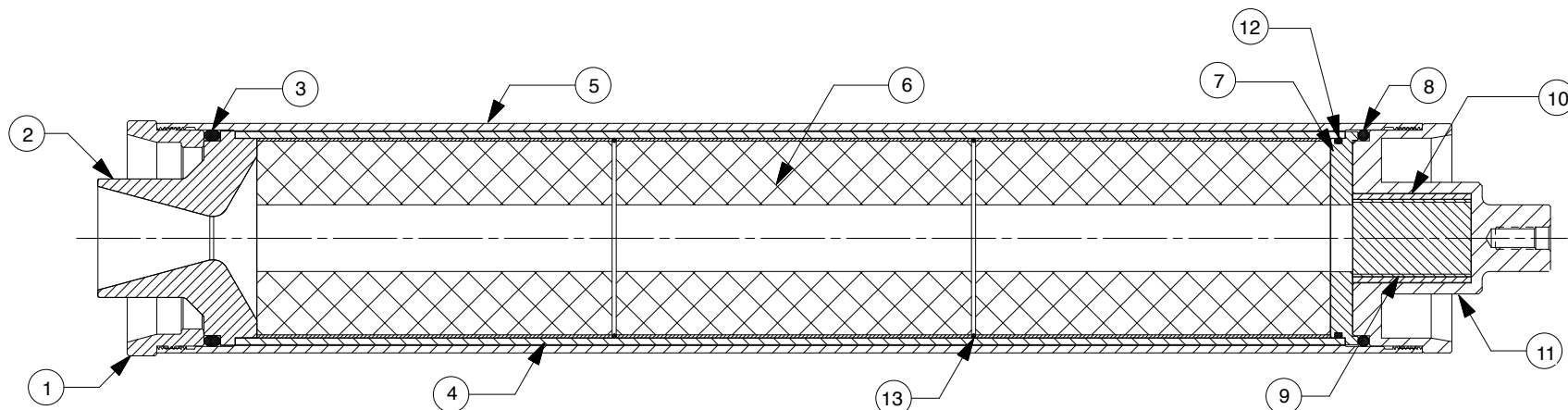
Motor Designation, Hardware and Typical Performance Data



RMS-98/2560-20480 Assembly Drawing and Instructions

NOTE: THE DRAWING SHOWN BELOW IS A GENERIC REPRESENTATION OF THE ACTUAL MOTOR. NOZZLE SIZE AND NUMBER AND DIMENSIONS OF PROPELLANT GRAINS MAY BE DIFFERENT.

ITEM	DESCRIPTION	QTY	ITEM	DESCRIPTION	QTY
1	AFT CLOSURE	1	10	SMOKE GRAIN INSULATOR	1
2	NOZZLE	1	11	FORWARD CLOSURE	1
3	AFT O-RING	1	12	FORWARD SEAL DISK O-RING	1
4	LINER	1	13	GRAIN SPACER O-RINGS	VARIES
5	CASE	1			
6	PROPELLANT GRAINS	VARIES			
7	FORWARD SEAL DISK	1			
8	FORWARD O-RING	1			
9	SMOKE GRAIN	1			



Assembly Instructions (numbers refer to item numbers on drawing):

1. Lightly grease forward, aft and forward seal disk o-rings (3, 8 & 12) and case threads (5).
2. Assemble smoke grain (9) and smoke insulator (10) as shown.
3. Apply a layer of grease to one end of the smoke grain and then push step 2 smoke assembly into forward closure (11), greased end first.
4. Install forward seal disk o-ring (12) on groove in forward seal disk (7) and insert this assembly into one end of liner (4) until seated.
5. Insert propellant grains (6) into liner (4) with grain spacer o-rings if included (13) between each grain, then push liner assembly into case (5) until recessed equally from both ends of case. If supplied, position the grain marked "aft grain" at the aft/nozzle end of the case. **NOTE:** If supplied, follow the grain bonding instruction sheet for this step.
6. Install forward o-ring (8) into (forward) end of case (5) with the forward seal disk (7).
7. Thread forward closure (11) into the (forward) end of the case (5) with the forward o-ring (8) until seated.
8. Assemble nozzle (2), aft o-ring (3), and aft closure (1) into open (aft) end of case (5) until seated.

Forward & Aft O-rings (3 & 8)
3-5/8" x 3/16" 339

Grain Spacer O-rings (13)
3-3/8" x 1/16" 042

Forward Seal Disk O-ring (12)
3-3/16" x 3/32" 151